Maxwell Thomas Asselmeier

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Education

Georgia Institute of Technology — Atlanta, GA
Ph.D. in Robotics (NSF Graduate Research Fellow) — *Aug. 2021 – May 2027*

University of Illinois at Urbana-Champaign — Champaign, IL
B.S. in Mechanical Engineering (Highest Honors) — *Aug. 2017 – May 2021*

Professional Experience

Mitsubishi Electric Research Laboratories (MERL) — Cambridge, MA
Optimization and Robotics Intern — *May 2026 – Oct. 2026*

- **IROS 2026 Humanoid IKEA Assembly Challenge** [ ]: Led a 4-person team to 1st place (of 11), deploying foundation models on the Unitree G1 and NVIDIA Jetson Thor via TensorRT for furniture assembly.
- Fine-tuned the NVIDIA GR00T N1.7 VLA on 750 teleoperated demos via distributed training (FSDP) on 8 A100s.
- **VLA Failure Detection** [ ]: Established a teleoperation system with the PICO 4U headset and SONIC whole-body controller to collect 300 demonstrations for kitting, used to post-train GR00T.
- Designed a VLM orchestrator that triggers locomotion and manipulation recovery behaviors to re-position the Unitree G1 into GR00T's operating regime, increasing task success rate by 50% (patent pending).

Institute for Human and Machine Cognition (IHMC) — Pensacola, FL
Humanoid Behaviors Intern — *May 2021 – Aug. 2021*

- **Atlas Perception Engine** [ ]: Designed and fabricated a custom multi-modal perception system including LiDARs, cameras, and IMUs; integrated into on-robot software stack for the Boston Dynamics Atlas humanoid.

Research Experience

Georgia Institute of Technology — Atlanta, GA
Graduate Research Assistant (Advisors: Ye Zhao and Patricio Vela) — *Aug. 2021 – Present*

- **Humanoid Motor Control** [ ]: Leverage trajectory optimization with kinematics and full-body dynamics constraints via Python and MuJoCo to generate reference motions for contact-rich manipulation.
- Deploy a whole-body controller trained via reinforcement learning (RL) using imitation-style rewards in Isaac Lab with PyTorch to track Unitree G1 hand poses for door opening on hardware with 3 cm error.
- **QuadPiPS** [ ]: Deployed a full-stack navigation codebase in C++ with SLAM and state estimation on the Unitree Go2 with a payload of a 64-beam Ouster LiDAR, RealSense RGB-D camera, and ZED stereo camera.
- Integrated real-time control loops onto a mini PC onboard the Go2 to enable untethered deployment over terrain.
- Mentored a team of 10 undergraduate and MS researchers, with 4 co-authoring publications.

Carnegie Mellon University — Pittsburgh, PA
Undergraduate Research Assistant (Advisor: Howie Choset) — *May 2020 – May 2021*

- **RL for Co-Design** [ ]: Validated the performance of PPO and TD3 on manipulation benchmarks in PyBullet.
- Trained robot arm co-design policies via Soft Actor-Critic (SAC) in Python to increase reachability volume by 28%.

Publications

5. **M. Asselmeier** and A. Schperberg. Robust Humanoid Foundation Model via Task-Aware Failure Detection and Recovery. Under Review, *IEEE ICRA*. 2027.
4. **M. Asselmeier**, Y. Zhao, and P. A. Vela. QuadPiPS: A Perception-informed Footstep Planner for Quadrupeds With Semantic Affordance Prediction. *IEEE T-RO*. 2026.
3. **M. Asselmeier**, D. Ahuja, A. Zaro, A. Abuaish, Y. Zhao, and P. A. Vela. Dynamic Gap: Safe Gap-based Navigation in Dynamic Environments. *IEEE ICRA*. 2025.
2. **M. Asselmeier**, J. Ivanova, Z. Zhou, P. A. Vela, and Y. Zhao. Hierarchical Experience-informed Navigation for Multi-modal Quadrupedal Rebar Grid Traversal. *IEEE ICRA*. 2024.
1. B. Mishra, D. Calvert, B. Ortolano, **M. Asselmeier**, L. Fina, S. McCrory, H. E. Sevil, and R. Griffin. Perception Engine Using a Multi-Sensor Head to Enable High-level Humanoid Robot Behaviors. *IEEE ICRA*. 2022.

Skills

Languages	C, C++, Java, MATLAB, Python
Software	CUDA, Drake, Isaac Sim/Lab, JAX, MuJoCo, PyTorch, ROS2, Slurm, TensorFlow, TensorRT, Unix/Linux
Concepts	3D Perception, Computer Vision, Foundation Models, Large-scale Distributed Systems, Low-level Control, Manipulation, Reinforcement Learning, Robot Co-Design, Sensor Fusion, State Estimation